1. What is the Internet

👉 Simple meaning:
The internet is a **network of networks** where computers talk to each other.

Every device has an **IP address** (like a house address).

Example:

- You type: google.com
- Your system finds its **IP**
- Then connects to that server

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2. IP Address

👉 An IP address is a **unique identifier** for a device.

Example:

192.168.1.1
8.8.8.8

Types:

- **Public IP** → visible on internet
- **Private IP** → inside your WiFi network

🧪 Try this:

```
ipconfig   # Windows
ifconfig   # Linux
```

👉 You'll see your IP.

💡 **Why Pentesters Need This:**

- Targets are identified using IP
- Scanning tools (like Nmap later) use IP

3. Domain Name vs IP

👉 Humans use names → computers use IP

Example:

- google.com → easy for humans
- 142.250.x.x → real IP

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4. DNS (Domain Name System)

👉 DNS = **Internet's phonebook**

It converts:

google.com → IP address

🧪 Try:

nslookup google.com

💡 **Why Important:**

- During recon, attackers gather DNS info
- Subdomain discovery depends on DNS

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5. Ports (VERY IMPORTANT)

👉 A computer has **multiple services**, each runs on a port.

Think:

- IP = building
- Port = room number

Common Ports:

Port	Service
80	HTTP
443	HTTPS
22	SSH
21	FTP
53	DNS

💡 **Why Important:**

- Pentesters scan ports to find **entry points**

Example:

- Port 22 open → maybe SSH login possible
- Port 80 → website exists

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6. TCP vs UDP

👉 Two ways data travels

TCP (Reliable)

- Connection-based
- Used in:
  - HTTP
  - SSH

UDP (Fast)

- No connection
- Used in:

- DNS
- Streaming

💡 **Why Important:**

- Nmap scans TCP/UDP differently
- Some attacks target UDP services

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7. OSI Model (Simplified)

You don't need to memorize all—just understand flow:

Important Layers:

1. **Application** → HTTP, FTP
2. **Transport** → TCP/UDP
3. **Network** → IP
4. **Data Link** → MAC

💡 **Why Important:**

- Helps debug network issues
- Helps understand where attack happens

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8. Packet (Core Concept)

👉 Data travels in **small chunks called packets**

Each packet contains:

- Source IP
- Destination IP

- Data

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9. Packet Analysis Tool

🔧 **Wireshark**

👉 What it does:

- Captures network traffic
- Shows packets

🧪 **What to do:**

1. Open Wireshark
2. Start capture
3. Visit a website
4. Stop capture

👉 You'll see packets flowing

💡 **Why Important:**

- Used to analyze traffic
- Detect vulnerabilities

10. How a Website Loads (Full Flow)

When you open a site:

1. You type URL
2. DNS finds IP
3. Browser sends request
4. Server responds
5. Page loads

11. HTTP Basics

Example request:

GET / HTTP/1.1

Types:

- GET → fetch data
- POST → send data

Status Codes:

- 200 → OK
- 404 → Not Found
- 500 → Error

💡 **Why Important:**

- Web hacking depends on HTTP
- Tools like Burp Suite intercept requests

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12. Browser DevTools

👉 Open:

- Right click → Inspect → Network tab

You'll see:

- Requests
- Headers
- Cookies

13. Linux Basics (START)

👉 Hackers mostly use Linux

Commands:

```
ls    # list files
cd    # change directory
pwd   # current path
```

Example:

```
cd Desktop
ls
```

🛠 **Tool:**

🔧 **Kali Linux**

👉 Special OS for hacking tools

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14. Permissions (IMPORTANT)

Each file has:

- Read (r)
- Write (w)
- Execute (x)

Command:

```
chmod +x file.sh
```

15. Cybersecurity Basics

CIA Triad:

- Confidentiality
- Integrity
- Availability

Attack Types:

- Brute force
- Phishing
- SQL Injection (basic idea)